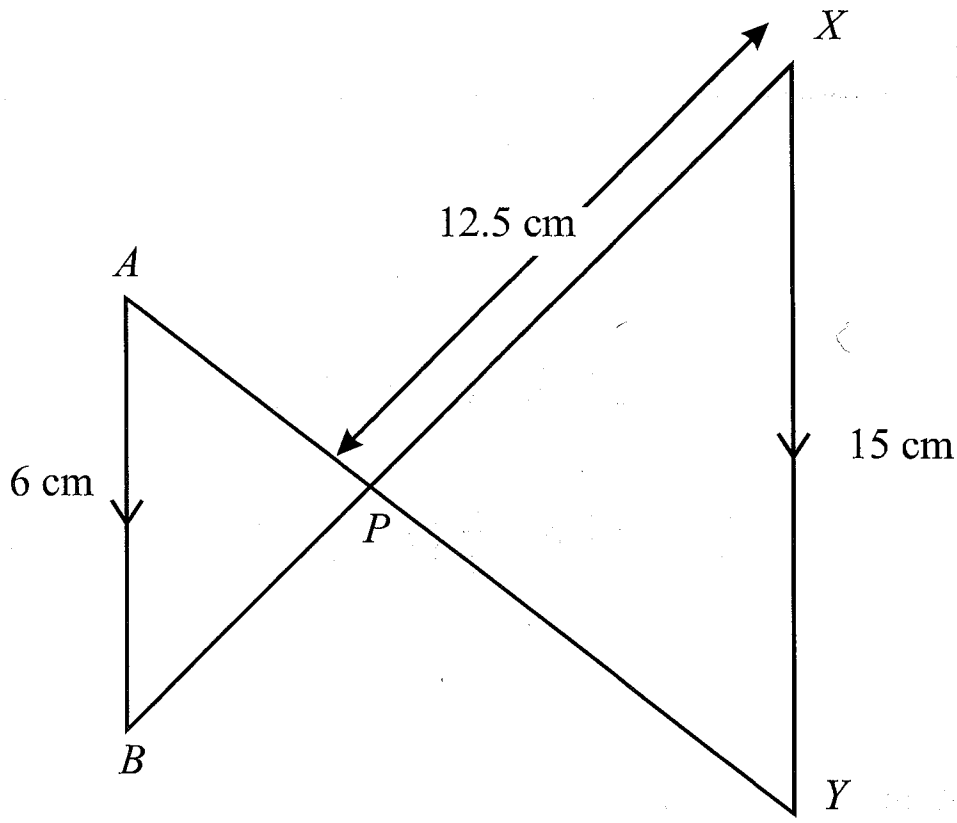
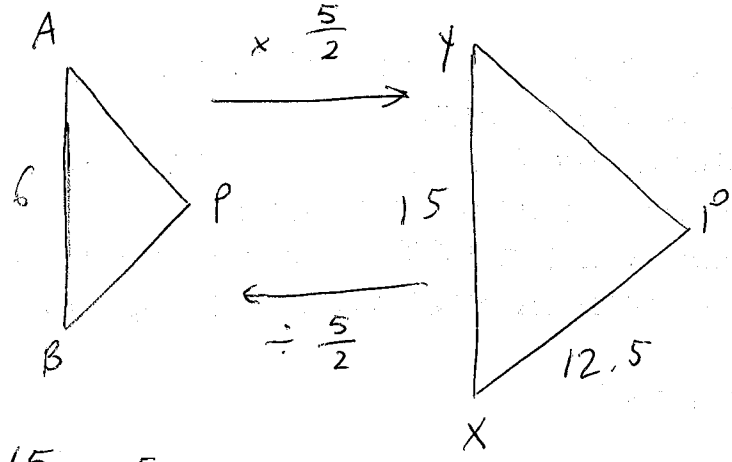
1



AB is parallel to *XY*.
The lines *AY* and *BX* intersect at *P*.
AB = 6 cm.
XP = 12.5 cm.
XY = 15 cm.

Work out the length of *BP*.



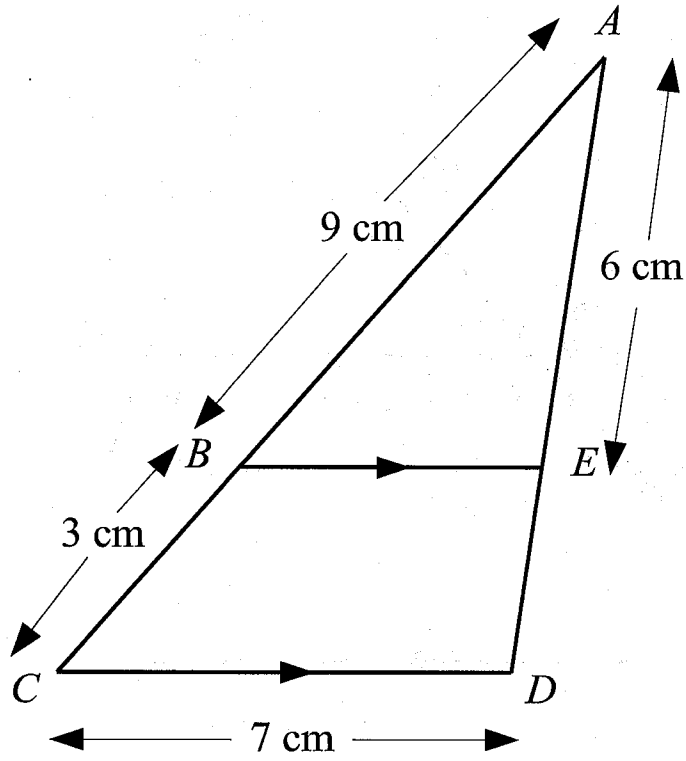
Scale factor $= \frac{15}{6} = \frac{5}{2}$

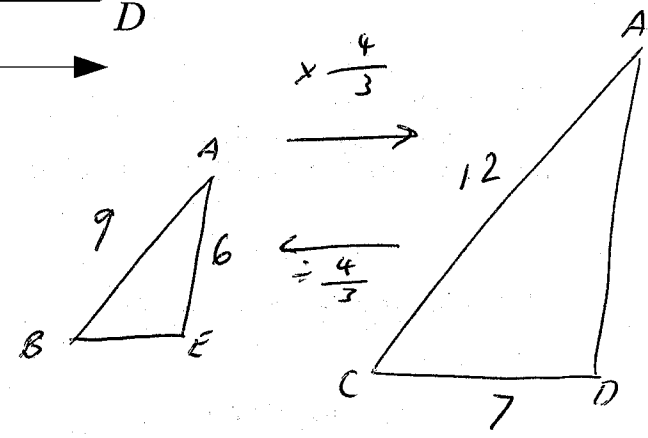
$12.5 \div \frac{5}{2} = \underline{\underline{5}}$

5 cm

(Total for Question 1 is 3 marks)

2





BE is parallel to *CD*.
AB = 9 cm, *BC* = 3 cm, *CD* = 7 cm, *AE* = 6 cm.

(a) Calculate the length of *ED*.

scale factor $= \frac{12}{9} = \frac{4}{3}$

$AD = 6 \times \frac{4}{3} = 8$ cm

$ED = 8 - 6 = 2$ cm

2 cm

(2)

(b) Calculate the length of *BE*.

$7 \div \frac{4}{3}$

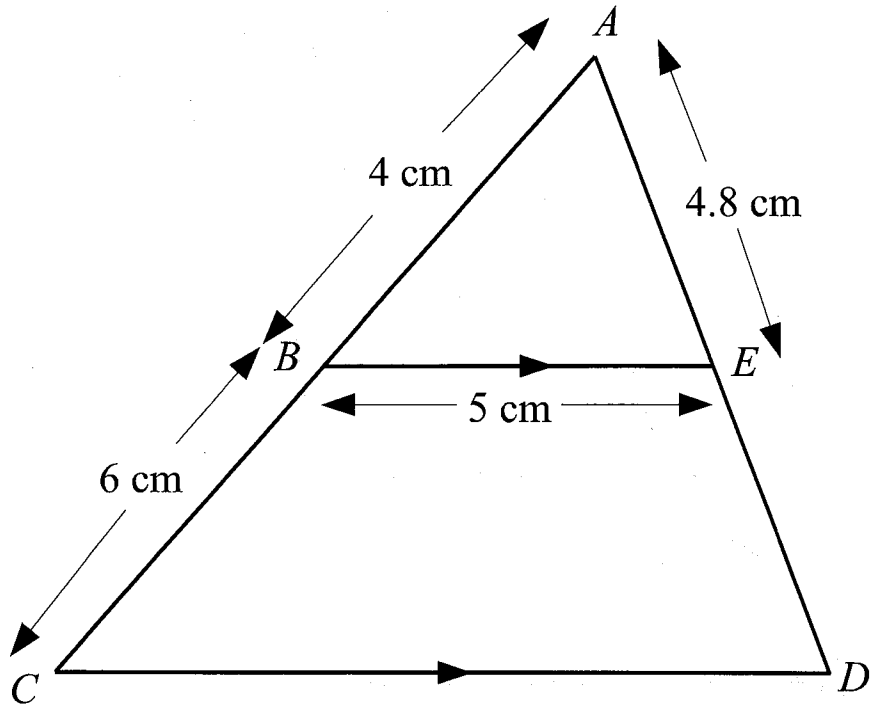
$7 \times \frac{3}{4} = \frac{21}{4}$ OR 5.25

$\frac{21}{4}$ cm

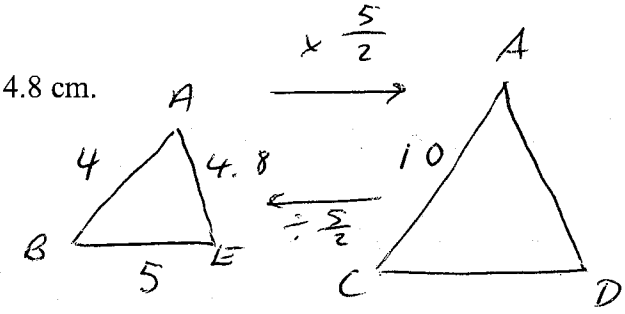
(2)

(Total for Question 2 is 4 marks)

3



BE is parallel to *CD*.
ABC and *AED* are straight lines.
AB = 4 cm, *BC* = 6 cm, *BE* = 5 cm, *AE* = 4.8 cm.



(a) Calculate the length of *CD*.

Scale factor $= \frac{10}{4} = \frac{5}{2}$

$5 \times \frac{5}{2} = \frac{25}{2}$ or 12.5

$\frac{25}{2}$ cm

(2)

(b) Calculate the length of *ED*.

$AD = 4.8 \times \frac{5}{2} = 12$ cm

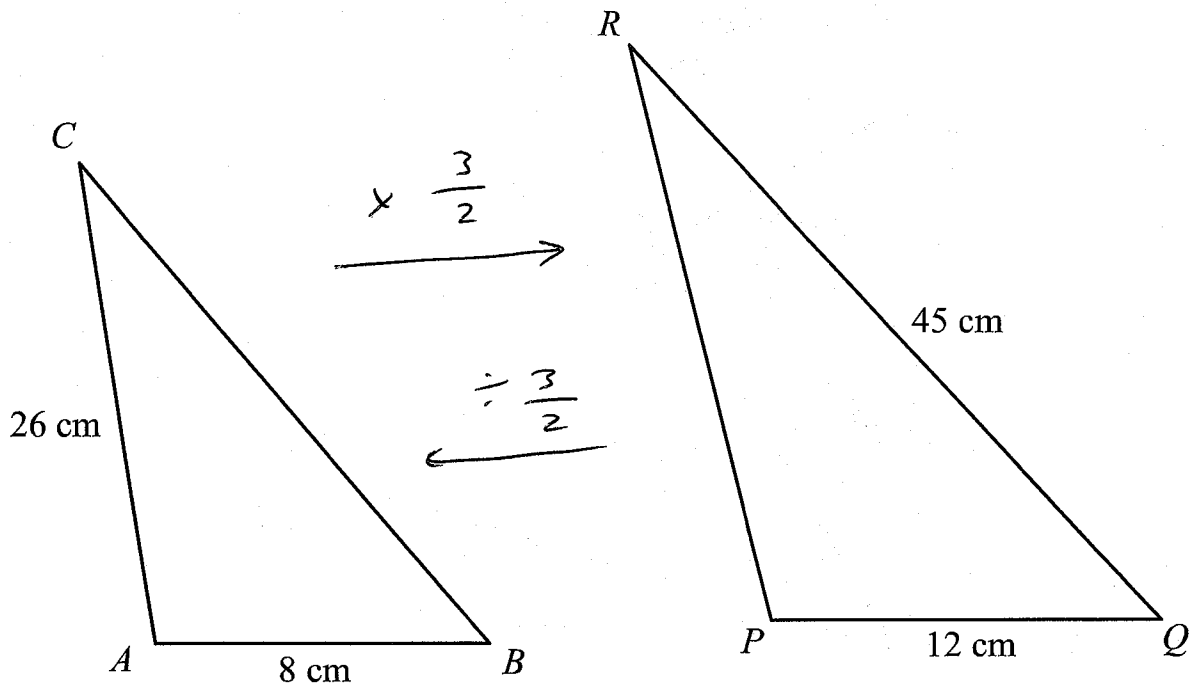
$12 - 4.8 = 7.2$ cm

7.2 cm

(2)

(Total for Question 3 is 4 marks)

4



The two triangles ABC and PQR are mathematically similar.
Angle A = angle P.
Angle B = angle Q.
AB = 8 cm.
AC = 26 cm.
PQ = 12 cm.
QR = 45 cm.

scale factor $= \frac{12}{8} = \frac{3}{2}$

(a) Calculate the length of *PR*.

$26 \times \frac{3}{2} = 39$

39 cm

(2)

(b) Calculate the length of *BC*.

$45 \div \frac{3}{2}$

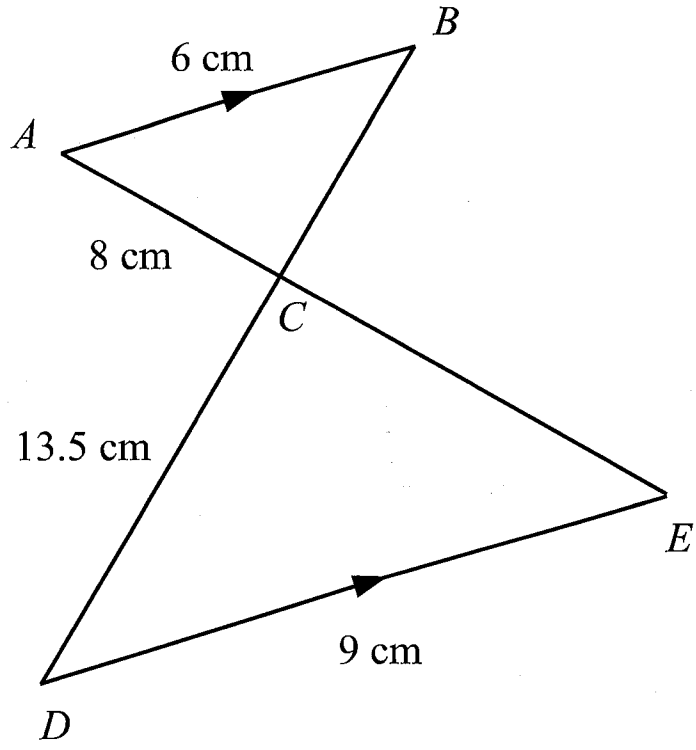
$45 \times \frac{2}{3} = 30$

30 cm

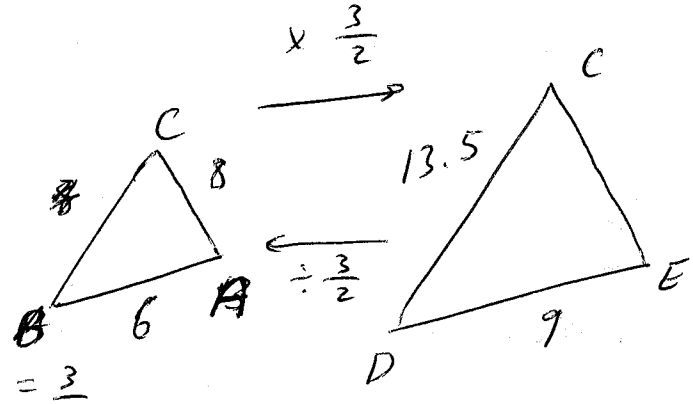
(2)

(Total for Question 4 is 4 marks)

5



AB is parallel to *DE*.
ACE and *BCD* are straight lines.
AB = 6 cm,
AC = 8 cm,
CD = 13.5 cm,
DE = 9 cm.



(a) Calculate the length of *CE*.

Scale factor $= \frac{9}{6} = \frac{3}{2}$

$8 \times \frac{3}{2} = 12$

12 cm

(2)

(b) Calculate the length of *BC*.

$13.5 \div \frac{3}{2}$

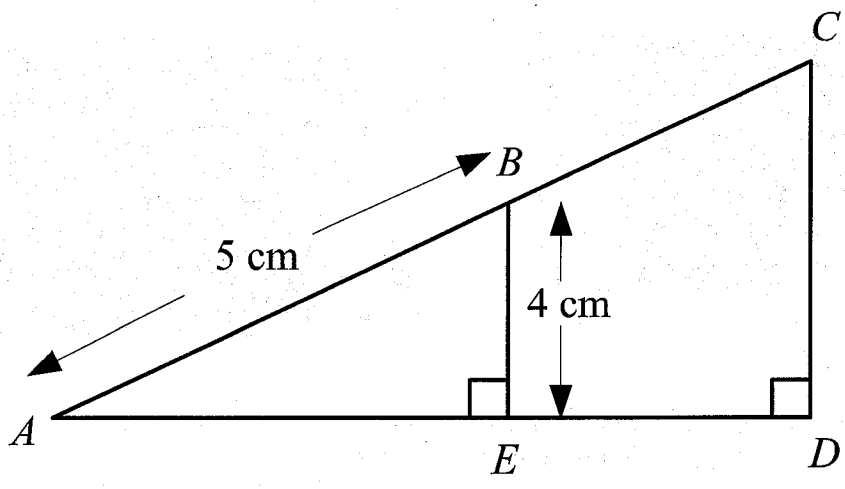
$13.5 \times \frac{2}{3} = 9$

9 cm

(2)

(Total for Question 5 is 4 marks)

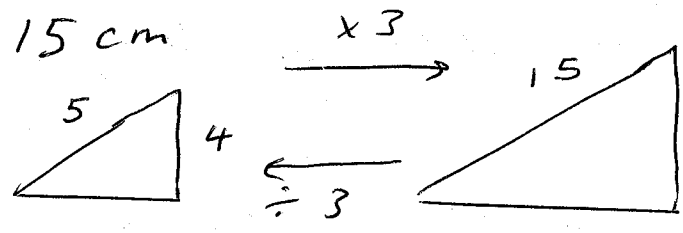
6



AC = 5 × 3

AB: AC = 1: 3

AC = 15 cm



(a) Calculate the length of *CD*.

Scale factor = $\frac{15}{5}$ = 3

4 × 3 = 12

12 cm

(2)

(b) Calculate the length of *BC*.

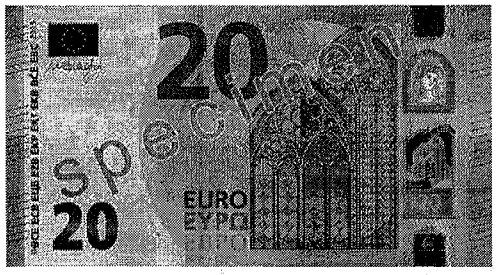
15 − 5 = 10

10 cm

(2)

(Total for Question 6 is 4 marks)

7



A 20 Euro note is a rectangle 133 mm long and 72 mm wide.
A 500 Euro Note is a rectangle 160 mm long and 82 mm wide.

Show that the two rectangles are not mathematically similar.

Scale factor for length $= \frac{160}{133}$

Scale factor for width $= \frac{82}{72} = \frac{41}{36}$

The scale factor for length <u>is not</u> equal to the scale factor for width $\therefore$ they are not similar.

(Total for Question 7 is 3 marks)